

Eversolar® UV-Curable Edge Sealing Architecture

Low-disruption upgrade path for PSC/Si Tandem module reliability

POE / TPO

Bulk encapsulation

AB-341k

Primary edge barrier

AB-402

Secondary structural seal

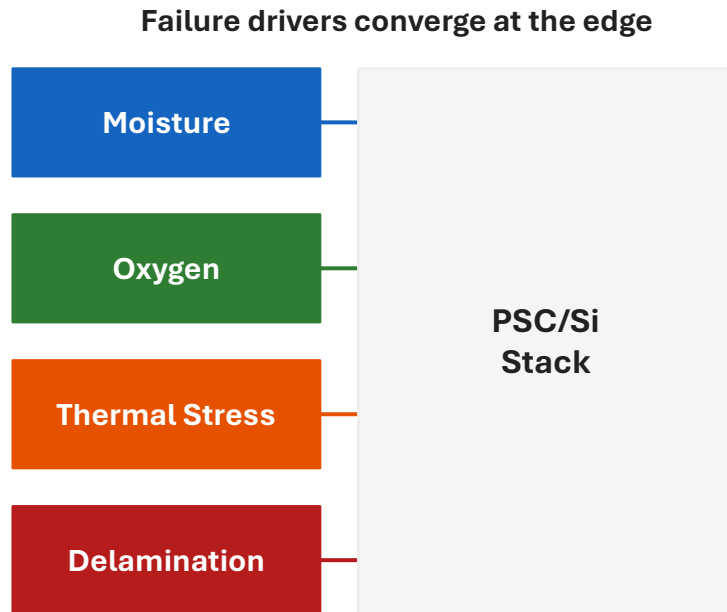
Keep the proven bulk encapsulation. Upgrade the edge.



1. Why Edge Sealing Is Critical for PSC/Si Tandem

PSC/Si Tandem has moved from cell efficiency to module reliability.

- Perovskite sub-cells are sensitive to moisture, oxygen and heat.
- Module edges are the primary lateral ingress pathway.
- Tandem stacks have more interfaces and higher thermal mismatch.
- Edge sealing is a reliability architecture element — not just a barrier.



The edge is where permeability, stress and interfacial durability all converge.

2. Conventional PV Edge Architecture

POE / TPO — Bulk encapsulation

- Proven lamination process
- High-throughput production
- Silicon PV baseline

PIB — Primary edge barrier

- Low moisture permeance
- Mature PV edge seal
- Industry benchmark

Silicone / PU — Secondary seal

- Mechanical integrity
- Prevents PIB creep/flow
- TC/DH compliance

Why it works for silicon

- Low moisture permeance at edge
- Mature materials and process
- Known module reliability baseline

What PSC/Si Tandem adds

- More sensitive perovskite sub-cell
- More interfaces, higher mismatch
- Edge/corner integrity more critical

The question: is the architecture designed for silicon still optimal for PSC/Si Tandem?

3. Initial Permeability Is Necessary — But Not Sufficient



For a moisture-sensitive device, a small edge defect can trigger failure — not just bulk permeability.

What literature shows	Implication for Eversolar
- PIB edge seals may flow — delamination can be catastrophic. - Ca film tests evaluate performance after thermal, UV and mechanical stress.	- AB-341k: benchmark on post-aging geometry retention — not only permeability. - AB-402: adds structural support and low-OTR secondary barrier.

1,930+h

ISOS-D3 T_{100}
AB-341 (Nature Comm. 2024)

1,000+h

ISOS-L2 T_{90}
AB-341 (EES Solar 2025)

0.0074

Permeability g·mm/m²·day
AB-341 (SGS certified)

0.00128

Permeability g·mm/m²·day
AB-341k (next-gen)

4. Commercial PIB Is the Reference Benchmark

H.B. Fuller HelioSeal® PVS-101 — Public Data

Property	Value
Chemical family	Polyisobutylene (PIB)
Product type	1K hot melt adhesive
Permeability	max. 0.234 g·mm/m ² ·day @40°C, 90%RH
Processing temp.	110 to 140°C
Tensile strength	min. 0.20 MPa
Mechanical note	Additional bonding recommended under high stress

A Balanced View



PIB remains the reference for edge protection in commercial silicon PV.



For PSC/Si Tandem, evaluation must extend beyond initial permeability to creep resistance and aging adhesion.



Mechanical integrity and adhesion durability after TC/DH become key differentiators.

PIB is not the problem. The opportunity is to upgrade the edge architecture for tandem modules.

5. Eversolar Proposal: Low-Disruption Edge Upgrade

Keep bulk lamination unchanged. Upgrade only the edge system.

Conventional Baseline		Eversolar Proposal	
POE / TPO	<i>Bulk encapsulation</i>	POE / TPO	<i>Bulk encapsulation (unchanged)</i>
PIB	<i>Primary edge barrier</i>	AB-341k	<i>Primary edge barrier (UV-curable)</i>
Silicone / PU	<i>Secondary structural seal</i>	AB-402	<i>Secondary structural seal (UV-curable)</i>

→ Edge changes

Why our partners can accept this path:

- ✓ No immediate replacement of POE/TPO lamination process.
- ✓ Lower requalification burden than full encapsulation stack redesign.
- ✓ Direct A/B validation against existing PIB + silicone edge system.
- ✓ Compatible with mini-module trial strategy.

6. AB-341k — UV-Curable Primary Edge Barrier

Target role: replace or benchmark against PIB as the primary edge moisture barrier.



Barrier

Low lateral moisture ingress.
SGS-certified Permeability by third-party lab.



UV Cure

Room-temperature dispensing and cure.
No hot-melt press required.



Geometry

Crosslinked network resists creep vs. uncrosslinked PIB.



Reliability

Designed for TC / DH / UV aging tests, not just initial permeability.

What AB-341k Must Prove

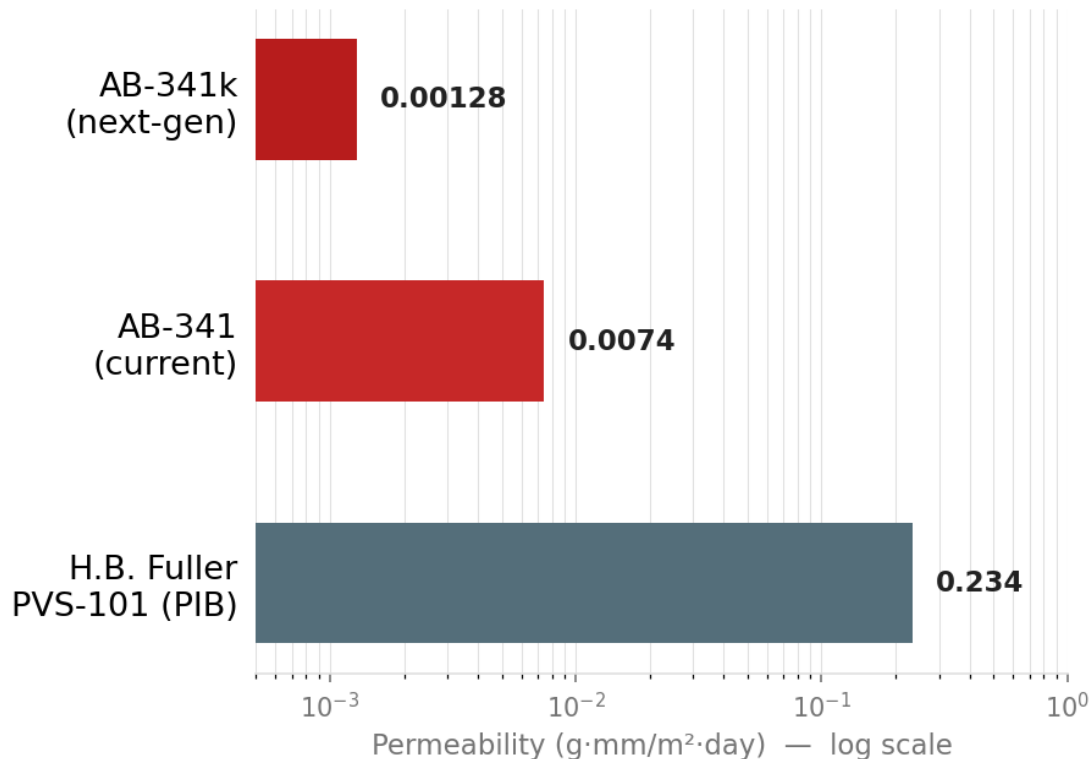
- Permeability comparable to PIB under matched conditions.
- Post-aging barrier retention (DH, UV, TC).
- Lower creep / better geometry retention at 85°C.
- Adhesion retention to glass and POE/TPO after aging.

Customer Message

AB-341k is positioned as a mechanically stable primary edge barrier — not merely a lower-permeability material.

The crosslinked network provides edge geometry fixation that uncrosslinked PIB cannot offer.

7. Permeability Comparison: AB-341k vs. AB-341 vs. PIB



Key Takeaways

~32×

AB-341 lower
than PIB
(0.0074 vs 0.234)

~183×

AB-341k lower
than PIB
(0.00128 vs 0.234)

~6×

AB-341k lower
than AB-341
(0.00128 vs 0.0074)

Unit: Permeability g·mm/m²·day @40°C, 90%RH. AB-341 & AB-341k: MOCON/SGS certified. PIB = H.B. Fuller PVS-101 public data.

8. AB-402 — Barrier-Enhanced Secondary Structural Seal

Target role: UV-curable alternative to conventional silicone secondary sealant.

Edge Function Stack

Glass substrate

AB-402 *Structural seal*

AB-341k *Primary barrier*

POE / TPO *Bulk encapsulation*

Glass substrate

AB-402 Value Hypothesis

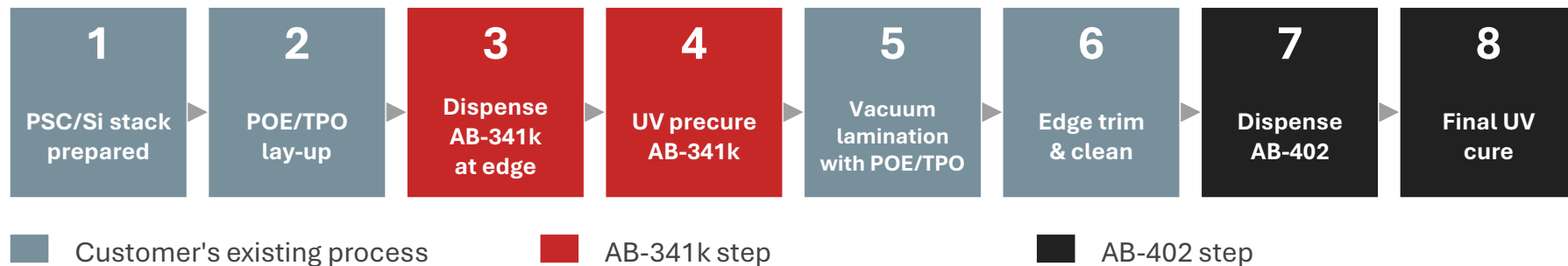
- Secondary structural support at module perimeter.
- Low OTR — additional oxygen barrier function.
- UV-curable: compatible with AB-341k workflow.
- Potential alternative to conventional silicone.

Proof Points Needed

- OTR vs silicone baseline
- Tensile / elongation / modulus
- UV / DH / TC aging durability

9. Proposed Process Flow: Hybrid Packaging

AB-341k replaces PIB before lamination; AB-402 remains the post lamination secondary seal.



Why this flow aligns with PIB based packaging

- PIB enters before lamination at the glass edge.
- AB-341k uses the same entry point plus UV precure.
- Bulk POE/TPO lamination concept remains unchanged.
- AB-402 remains the post lamination secondary seal.

Process risks to validate

- Geometry retention after heat press.
- Glass adhesion and POE/TPO interface.
- Void, overflow and edge gap control.
- Lamination temperature / pressure window.

10. Validation Plan: A/B Architecture Comparison

The proof must be module-level edge integrity after standardized stress.

Control A

POE/TPO + commercial PIB + silicone

Test B

POE/TPO + AB-341k + AB-402

Stress Tests

DH

85°C/85%RH

TC

TC200/TC400

UV

UV aging

HF

Humidity freeze

Outdoor

Field exposure

Comparison Matrix

Category	Key Metrics
Barrier	Permeability, OTR, post-aging retention
Edge integrity	Ca ingress imaging, cross-section, delamination mapping
Mechanics	Creep, adhesion retention, TC edge integrity
Device relevance	PCE retention, edge darkening, corrosion signs

Traditional PV encapsulation has been optimized for silicon modules.

PSC/Si Tandem introduces additional sensitivity to moisture, oxygen, temperature and edge-driven degradation.

Eversolar AB-341k + AB-402 is a UV-curable edge sealing architecture that keeps our partner's existing POE/TPO bulk encapsulation, while upgrading the primary barrier and secondary structural seal.

Suggested Next Step

Run an A/B mini-module validation against your current PIB + silicone edge system.

References

Public literature and product data used in this discussion deck.

1

De Bastiani et al., "All Set for Efficient and Reliable Perovskite/Silicon Tandem Photovoltaic Modules?" Solar RRL, 2021.

2

Dameron, Reese, Kempe et al., "Evaluation and Modeling of Edge-Seal Materials for Photovoltaic Applications," NREL/IEEE PVSC, 2010.

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Kempe et al., "Modeling of the Effectiveness of Novel Edge Seal Designs," Energy Science & Engineering, 2023.

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H.B. Fuller HelioSeal® PVS-101 public product data. <https://www.hbfullerproducts.com/products/helioseal-pvs-101>

6

Gallant et al., Nature Communications 2024. ISOS-D3 $T_{100} > 1930\text{h}$ with Eversolar AB-341 encapsulation.

7

Gomar-Fernández et al., EES Solar 2025. ISOS-L2 $T_{90} > 1000\text{h}$ with Eversolar AB-341 encapsulation.